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function ss2_problem1(n)
% Lab 1: Sampling Theorem Reconstruction
% n: input parameter controlling the sampling frequency

% --- (b) Define signal parameters ---
T = 0.0025;           % Base temporal resolution
t = 0.0025:T:1;       % Time interval t from [0.0025, 1]

% The signal:  $x(t) = 4\sin(2\pi t) + \cos(\pi/4 + 16\pi t)$ 
x = 4*sin(2*pi*t) + cos(pi/4 + 16*pi*t);

% --- (c) Sampling the signal ---
Ts = T * n;           % Sampling interval
tk = downsample(t, n); % Sampled time points
xk = downsample(x, n); % Sampled signal values

% --- (e) Signal Reconstruction ---
% Initialize reconstructed signal vector
x_recon = zeros(size(t));

% Reconstruction formula: sum of  $x_k * \text{sinc}((t - kTs)/Ts)$ 
% Note: MATLAB's sinc function is defined as  $\sin(\pi x)/(\pi x)$ 
for k = 1:length(xk)
    x_recon = x_recon + xk(k) * sinc((t - tk(k)) / Ts);
end

% --- Plotting ---
figure('Name', ['Sampling Analysis for n = ', num2str(n)]);

% First Subplot: Original and Sampled
subplot(2,1,1)
plot(t, x, 'b', 'LineWidth', 1.2); hold on;
stem(tk, xk, 'r', 'filled', 'LineWidth', 1.0);
xlabel('t');
ylabel('Amplitude');
title(['Original and Sampled Signals (n = ', num2str(n), ')']);
legend('Original x(t)', 'Samples x_k');
grid on;
xlim([0 1]);

% Second Subplot: Reconstructed Signal
subplot(2,1,2)
plot(t, x, 'b:', 'LineWidth', 1.0); hold on; % Original for reference
plot(t, x_recon, 'r', 'LineWidth', 1.2);
xlabel('t');
ylabel('Amplitude');
title('Reconstructed Signal  $\tilde{x}(t)$ ');
legend('Original (Ref)', 'Reconstructed');
grid on;
xlim([0 1]);

% Ensure vertical axes are identical for comparison

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    linkaxes(findall(gcf, 'type', 'axes'), 'xy');  
end  
  
Not enough input arguments.  
  
Error in ss2_problem1 (line 13)  
    Ts = T * n;           % Sampling interval  
        ^
```

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